

Score based generative modeling

Geometric Data Analysis Project Presentation



Kshitij Ambilduke



Nemanja Vujadinovic



Bruny Soutarson

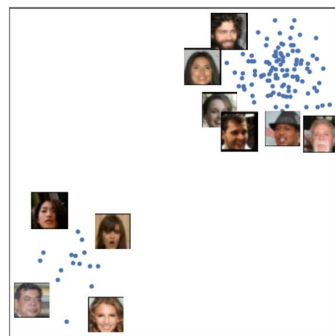
Content

- Introduction
- Limitations
- Discrete variance schedule
- Continuous variance schedule
- Latent generation

Introduction

Estimation of Non-Normalized Statistical Models by Score Matching

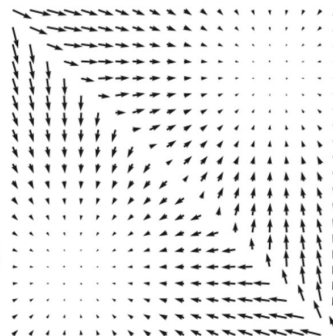
Aapo Hyvärinen



Data samples

$$\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \stackrel{\text{i.i.d.}}{\sim} p(\mathbf{x})$$

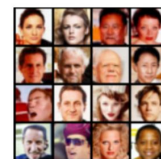
score
matching



Scores

$$\mathbf{s}_\theta(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

Langevin
dynamics



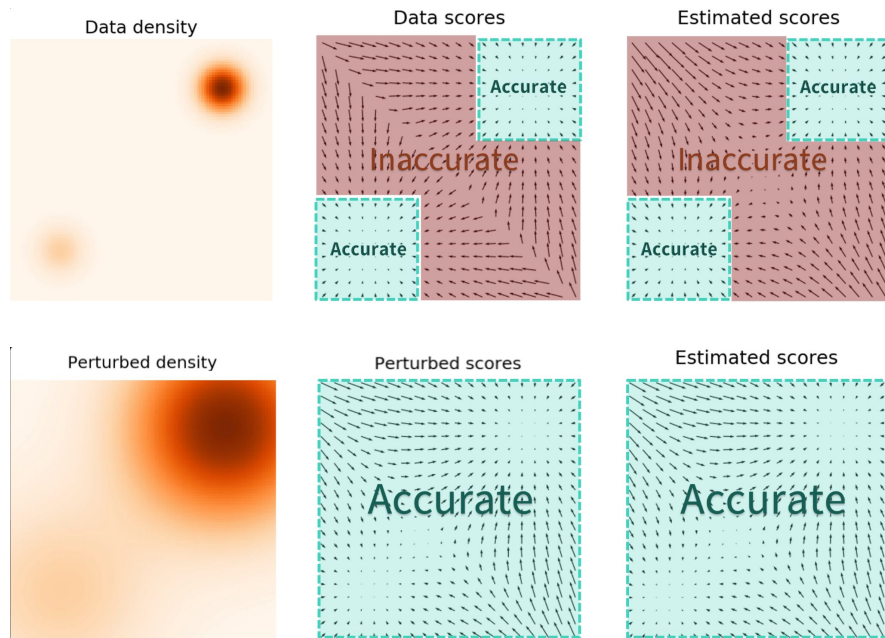
New samples

A Connection Between Score Matching and Denoising Autoencoders

Pascal Vincent

Images from Y. Song blog [2]

Introduction



Generative Modeling by Estimating Gradients of the Data Distribution

Yang Song
Stanford University
yangsong@cs.stanford.edu

Stefano Ermon
Stanford University
ermon@cs.stanford.edu

Limitations

About scores:

- Score is undefined when data is confined to low dimensional manifold
- Estimated score not accurate for regions with low data density

Euclidean-only formulation:

- Training/sampling assumes flat vector spaces
- Non-trivial to extend to Riemannian manifolds
- Existing Riemannian methods are not yet scaled as effectively [1][2]

Slow sampling:

- Requires many Langevin steps
- Much slower than VAEs or GANs
- Limits real-time use/large-scale data generation

[1] De Bortoli *et al.* *Riemannian Score-Based Generative Modelling*, 2022.

[2] Huang *et al.* *Riemannian Diffusion Models*, 2022.

Discrete variance schedule

Geometric schedule: $\sigma_i = \sigma_1 \left(\frac{\sigma_L}{\sigma_1} \right)^{\frac{i-1}{L-1}}$, $i = 1, \dots, L$.

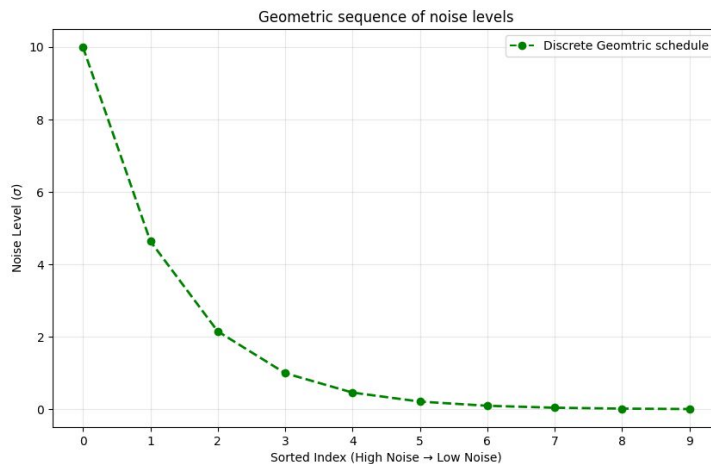
Step size: $\alpha_i = \epsilon \cdot \frac{\sigma_i^2}{\sigma_L^2}$

Intuition:

- High $\sigma_i \Rightarrow$ large $\alpha_i \Rightarrow$ fast exploration
- Low $\sigma_i \Rightarrow$ small $\alpha_i \Rightarrow$ refinement near data modes

Questions [1]:

- Is the decrease too aggressive?
- Are fine details emphasized before global structure?



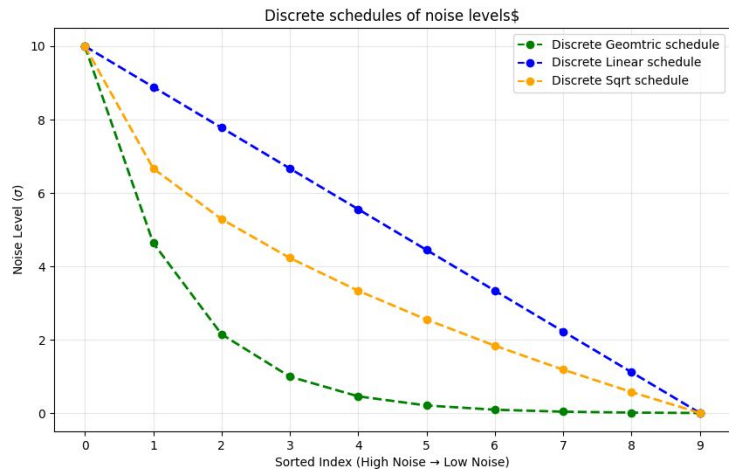
Discrete variance schedule

Linear schedule: $\sigma_i = \sigma_1 + (\sigma_L - \sigma_1) \frac{i-1}{L-1}$

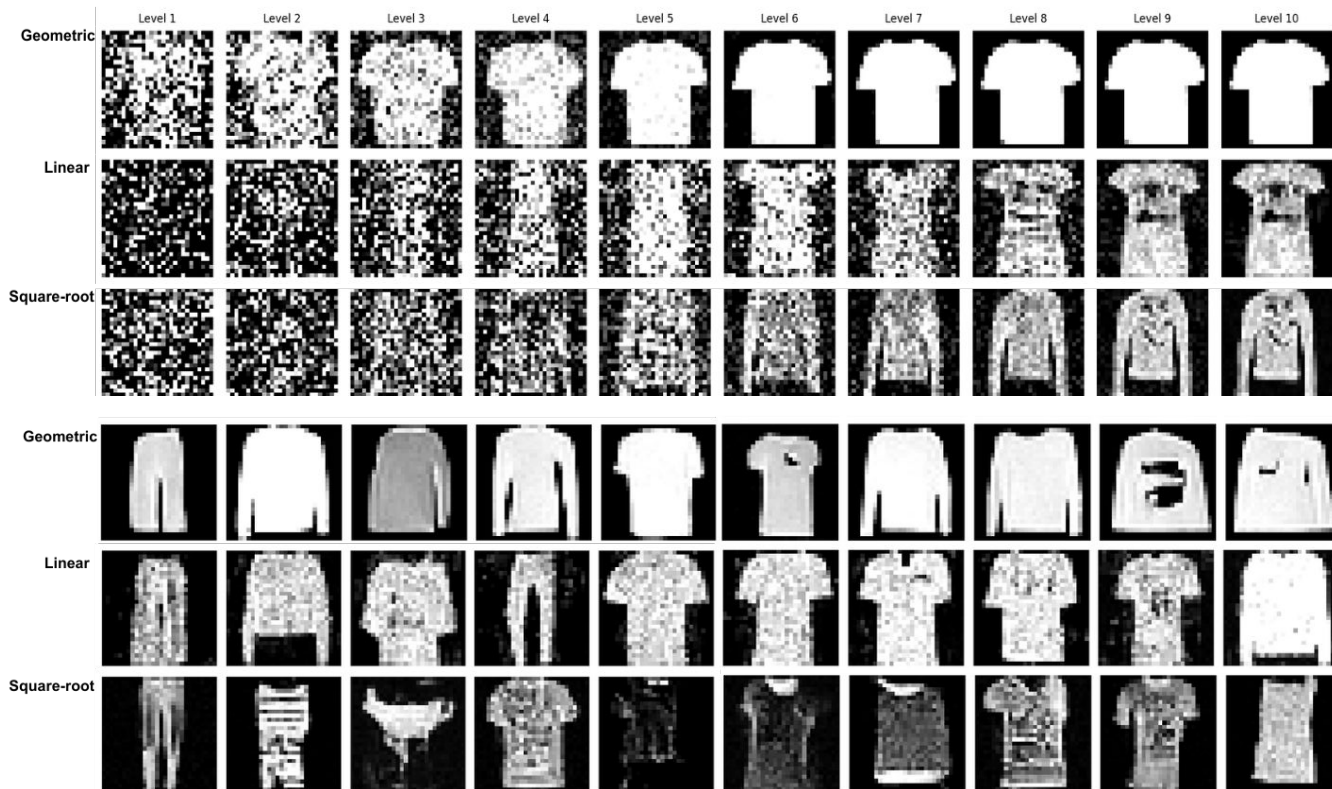
Square-root schedule: $\sigma_i = \sigma_1 + (\sigma_L - \sigma_1) \sqrt{\frac{i-1}{L-1}}$

Hypothesis:

- Linear: Better global structure sampling
- Square-root: Balance between other two



Discrete variance schedule



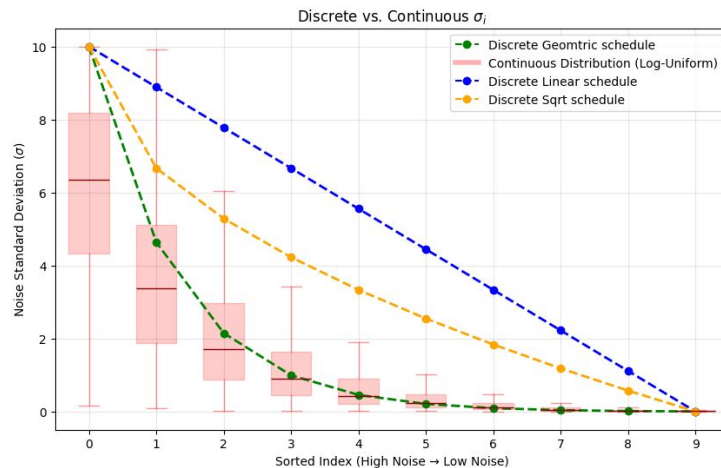
Continuous variance schedule

Training

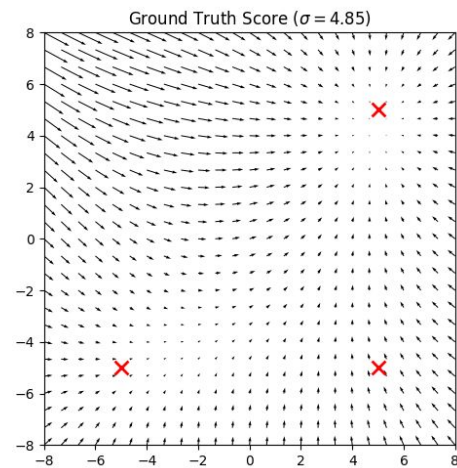
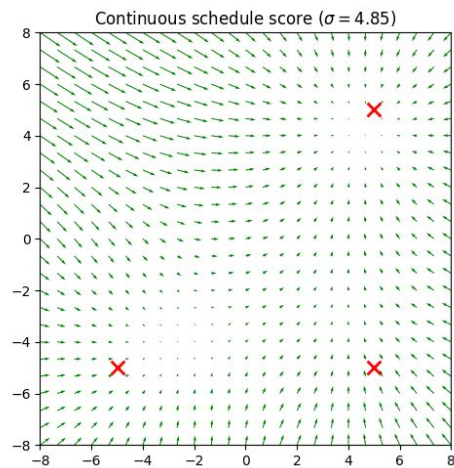
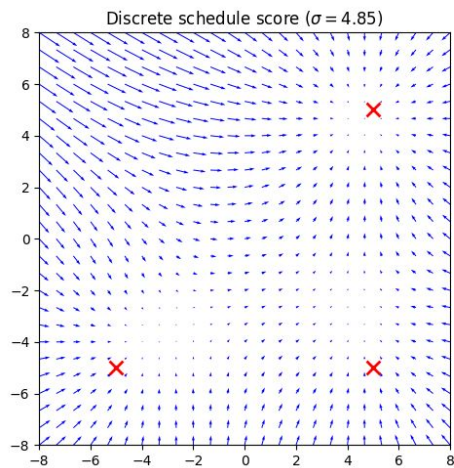
$$\sigma = \sigma_1 \cdot \left(\frac{\sigma_L}{\sigma_1} \right)^u, \quad \text{where } u \sim \mathcal{U}(0, 1).$$

Sampling: Interpolation the log space

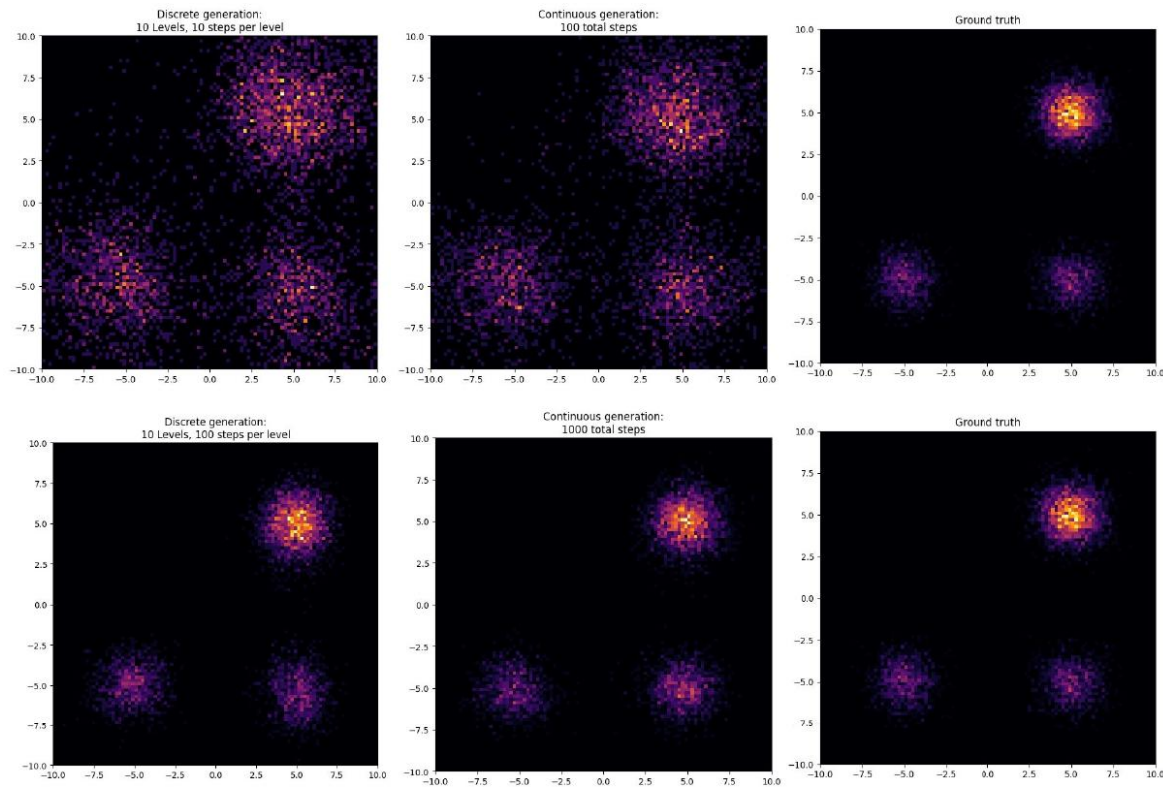
$$\begin{aligned} \log(\sigma_t) &= (1 - t) \cdot \log(\sigma_1) + t \cdot \sigma_L \\ \sigma_t &= \exp(\log(\sigma_t)) \end{aligned}$$



Continuous variance schedule



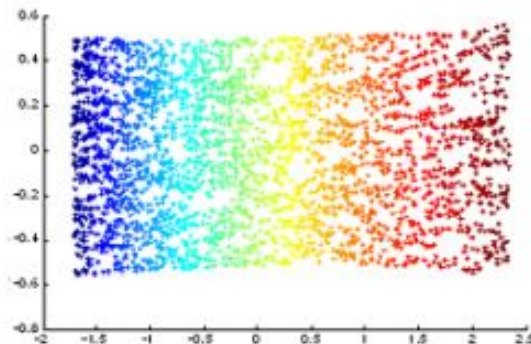
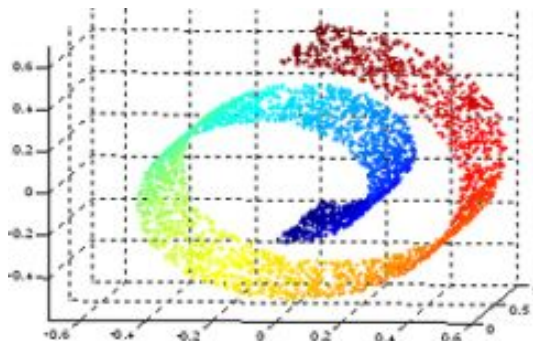
Continuous variance schedule



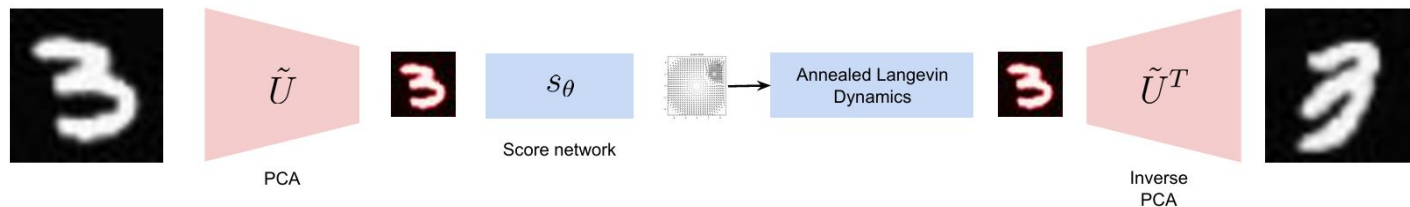
Latent generation

Recap of limitations

- Estimated score not accurate for regions with low data density
- Sampling is slower as compared to GANs or VAEs



Latent generation



Training

- Encode the input image into a lower dimensional latent using the encoder
- Train the model to predict scores of the latent images

Sampling

- Start from pure noise in the latent space.
- Generate a novel latents using Annealed Langevin Dynamics.
- Decode the generated latent back to the image space using the decoder

Latent generation

Latent space has smaller dimension !

Faster training

(Measured on T4 GPU)

Data space: 617 sec/epoch

Latent space: 210 sec/epoch

~ 66% decrease

Faster inference

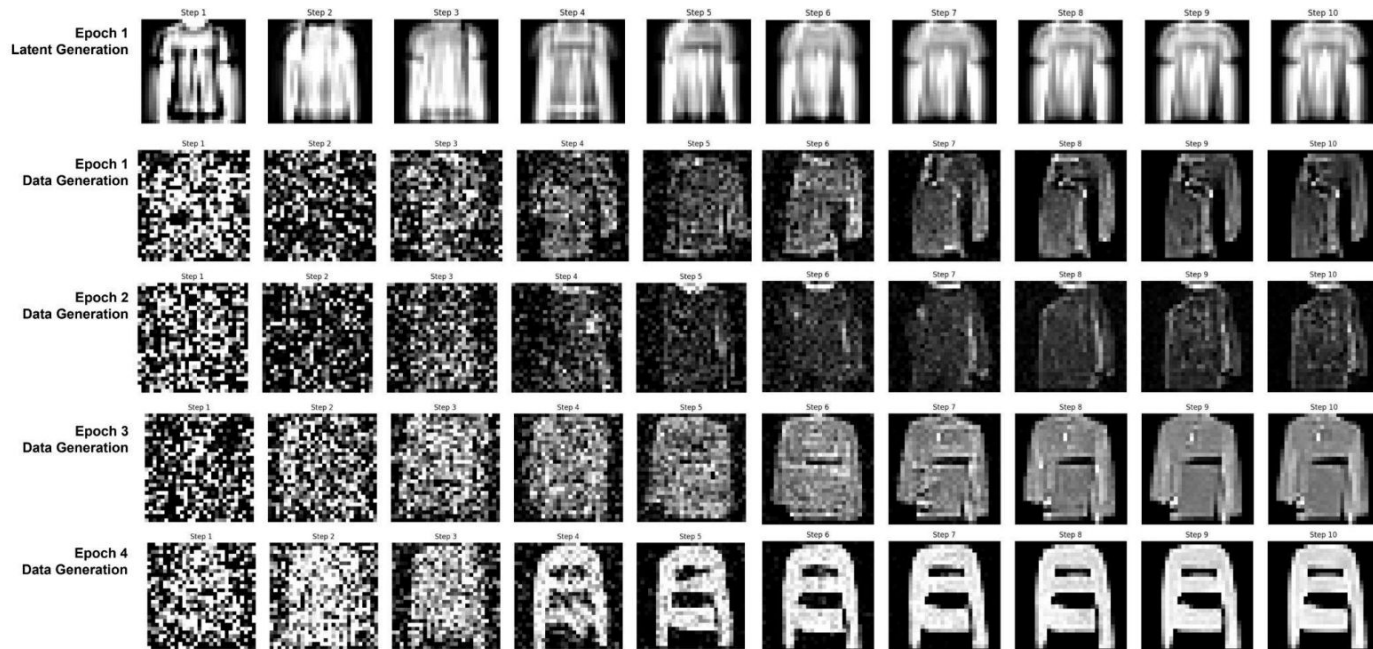
(Measured on NVIDIA GTX 1650)

Data space: 68 sec/epoch

Latent space: 27 sec/epoch

~ 60% decrease

Latent generation



Latent generation

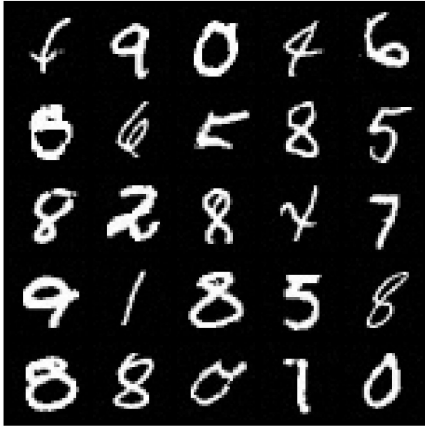


Figure 10: MNIST Digits generated by the classical approach.

Sample A		Sample B	FID Score
MNIST Dataset	Raw data generations		41.34
Recon. MNIST Dataset	Latent generations		52.08
MNIST Dataset	Latent generations		63.49

Table 2: FID scores on MNIST



Figure 9: MNIST Digits generated by our latent PCA model.

Sample		Inception score
MNIST Dataset		2.466 ± 0.04
Raw data generations		2.351 ± 0.04
Latent generations		2.232 ± 0.012

Table 1: Inception scores for MNIST and generated samples

Conclusion

Qualitative reproduction

- Visually plausible results for MNIST and FashionMNIST

Continuous variance schedule analysis

- Improved generalization to unseen schedule levels
- Enhanced sampled quality under limited sampling budgets

Discrete variance schedule analysis

- Insights into linear and square-root schedules

Latent space generation

- Reduced training and sampling time with faster convergence

THANK YOU!